Deep Recovery of Effective Migration Surfaces

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Abstract

Spatial population genetic data often exhibit isolation by distance, but the rate at which genetic similarity decays can vary across a habitat. EEMS and FEEMS visualize this heterogeneity as a surface of relative effective migration. FEEMS obtains a surface by penalized optimization and selects its spatial regularization separately for each dataset, which can require many fits. Here we introduce Deep Recovery of Effective Migration Surfaces (DREEMS), a simulation-trained graph neural network that estimates relative migration from genetic covariance and a geographic graph. DREEMS uses graph signals at several spatial scales, an approximate resistance inversion, and a short covariance refinement. On 512 independently simulated landscapes, DREEMS attained mean edge correlation 0.720 and mean squared error 0.533, compared with 0.614 and 0.738 for FEEMS using documented joint leave-one-observed-deme-out cross-validation. DREEMS had higher correlation in 81.8% and lower error in 83.8% of examples. Nominal 95% predictive intervals covered 95.7% of edges in an independent calibration experiment. Median complete runtime was 8.4 seconds for DREEMS and 151.6 seconds for FEEMS in the executed workflows. We also applied one checkpoint to 15 public datasets. In North American gray wolves, FEEMS fit covariance better and recovered a stronger Hudson Bay barrier. DREEMS provides a fast exploratory estimate of spatial structure, while its simulation dependence makes model checking important for empirical use.

1 Introduction

The relationship between geography and genetic variation has long been a central problem in population genetics. Individuals sampled near one another tend to be more genetically similar than individuals sampled far apart, a pattern commonly described as isolation by distance (Wright, 1943; Rousset, 1997). This relationship is rarely uniform across a species' range. Coastlines, mountain ranges, habitat transitions, and historical events can make the same geographic distance correspond to very different levels of genetic differentiation.

EEMS was developed to visualize spatially heterogeneous isolation by distance without assigning samples to discrete populations (Petkova et al., 2016). It represents a habitat by a dense stepping-stone graph and estimates relative effective migration across the graph. The model has an electrical interpretation: migration rates act as conductances, and effective resistance approximates genetic dissimilarity (McRae, 2006). Regions of low effective migration can reveal barriers to gene flow, whereas regions of high effective migration can reveal unusually strong connectivity. These are effective summaries of spatial genetic structure, not direct estimates of contemporary dispersal.

Fast Estimation of Effective Migration Surfaces (FEEMS) placed this idea in a penalized-likelihood framework and made larger graphs practical (Marcus et al., 2021). It estimates a separate parameter for each graph edge, which allows flexible and anisotropic surfaces, and uses spatial regularization to control overfitting. The amount of regularization is important: a strong penalty gives an almost uniform surface, while a weak penalty can produce a patchy fit. The documented

FEEMS procedure chooses migration and local-drift penalties by leaving out each observed deme in turn. This is statistically appealing, but it requires many optimizations for every new dataset.

Here we consider whether these repeated fits can be amortized. A model trained on many simulated inverse problems can learn how patterns in genetic covariance map to plausible conductance surfaces, then reuse that information for a new dataset. This approach has been successful for other population genetic problems (Flagel et al., 2019; Smith et al., 2024). Its weakness is equally clear: the answer depends on the training distribution. A network can perform well on average and still fail when the graph, sampling design, or demographic history differs from its simulations.

We introduce Deep Recovery of Effective Migration Surfaces (DREEMS). The method takes a genetic covariance matrix, sample locations, and a geographic graph. It first computes a linearized approximation to the conductance inverse problem. It then represents covariance and sampling signals at several graph-diffusion scales and passes them through a graph neural network. The network predicts relative migration for every edge and a predictive standard deviation. A short, fixed refinement step improves the fit to the observed covariance while keeping the result near the neural estimate. The representation uses graph topology and relative edge length but no absolute geographic axis or environmental covariates.

We evaluate one fixed checkpoint on 512 simulated landscapes and compare it with the upstream FEEMS implementation using its documented leave-one-observed-deme-out procedure. We examine edge recovery, sampling density, graph size, runtime, and predictive uncertainty. We then apply the same checkpoint without retraining to 15 public datasets, including the North American gray wolves used to introduce FEEMS. The simulations show a broad average advantage for DREEMS, while the wolf analysis gives a concrete case in which the dataset-specific FEEMS fit remains more convincing.

2 Results

2.1 Overview of DREEMS

Figure 1 summarizes the DREEMS workflow. The inputs are a genetic covariance matrix, sample coordinates, and a graph whose nodes have geographic coordinates. Samples are assigned to nearby graph nodes and their covariance is aggregated by deme. The method does not receive coastlines, elevation, habitat labels, or a preliminary FEEMS estimate.

The first stage uses all pairs of observed demes to construct a linearized estimate of edge conductance. This calculation asks which graph edges carry electrical flow between each sampled pair and accumulates evidence from their genetic dissimilarity. It is deliberately approximate: its purpose is to give the network an informative starting field without fitting a full migration surface.

The second stage spreads covariance and sampling signals over the geographic graph for several diffusion times. Differences between successive smoothed signals form bands that retain local, regional, and broad-scale information. An edge-aware graph neural network combines these bands with the linearized field. Local message passing describes nearby structure, while a graph-wide attention branch allows each edge to use information from distant sampled regions. The output is a mean relative log-migration rate and predictive standard deviation for each edge, together with a node-level local-drift field.

Finally, a fixed 50-step optimization adjusts the neural surface to better match the observed covariance. The adjustment is restricted to an uncertainty-scaled neighborhood of the neural prediction. Unlike FEEMS cross-validation, this refinement has no dataset-specific tuning search. The same checkpoint and refinement settings are used for every analysis.

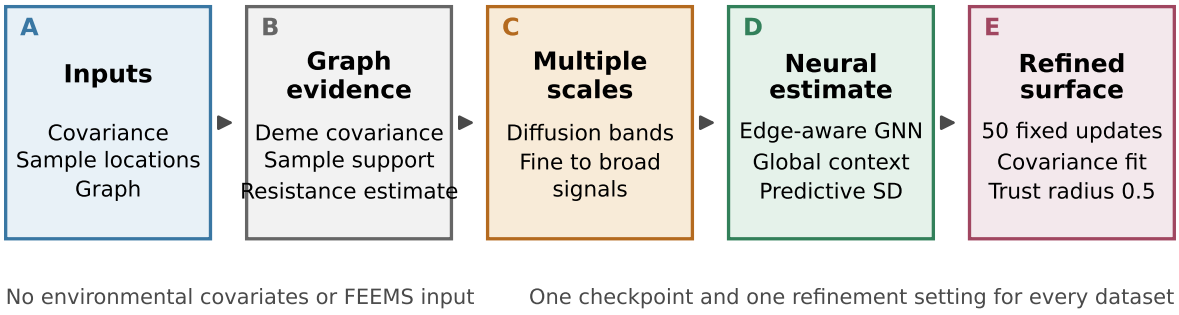


Figure 1: Overview of DREEMS. Genetic covariance and sample locations are placed on a user-supplied geographic graph. An all-pairs resistance approximation and fixed diffusion bands provide edge and node features to a graph neural network. The network predicts relative migration and uncertainty, and a fixed trust-region step refines the surface against observed covariance.

2.2 Training on simulated spatial histories

We trained DREEMS on 61,440 structured-coalescent simulations. The graphs contained 9–1,743 nodes and included regular lattices, masked habitats, corridors, and random windows sampled from landmasses around the world (Table 12). Migration surfaces included homogeneous isolation by distance, smooth heterogeneity, barriers, corridors, anisotropy, coast effects, and variation in local drift. Sampling ranged from nearly full coverage to a small fraction of graph nodes and was allowed to correlate with the underlying migration process. Landscape variables generated the simulation targets, but they were not model inputs.

Validation loss fell quickly during the first 10,000 updates and improved more slowly thereafter (Figure 2). The lowest graph-balanced validation loss occurred at step 29,000. Training stopped at step 41,000 after 12 further checks failed to improve the loss by the prespecified threshold, and the step-29,000 parameters were restored. The separate set of 4,096 held-out examples was not used to select the checkpoint.

2.3 DREEMS recovered simulated migration surfaces

We generated 512 independent global-landscape examples for the main comparison. These graphs contained 219–1,406 nodes and 590–4,065 edges. A median of 39 graph nodes was observed, corresponding to a median sampled-node fraction of 0.049. Each example contained a new graph window, migration surface, sampling design, and structured-coalescent genotype simulation.

We ran FEEMS on the same genotype matrix, coordinates, and graph. For each example, FEEMS jointly selected its migration and local-q smoothing parameters by documented leave-one-observed-deme-out cross-validation and then refit the selected model to all samples. In contrast, DREEMS used the same checkpoint and radius-0.5 covariance refinement for all 512 examples.

Refined DREEMS attained mean edge correlation 0.720 and mean squared error (MSE) 0.533, compared with correlation 0.614 and MSE 0.738 for FEEMS (Table 1). Before refinement, the neural estimate had mean correlation 0.698 and MSE 0.558. The fixed refinement therefore improved both metrics without using the simulated truth to choose its radius.

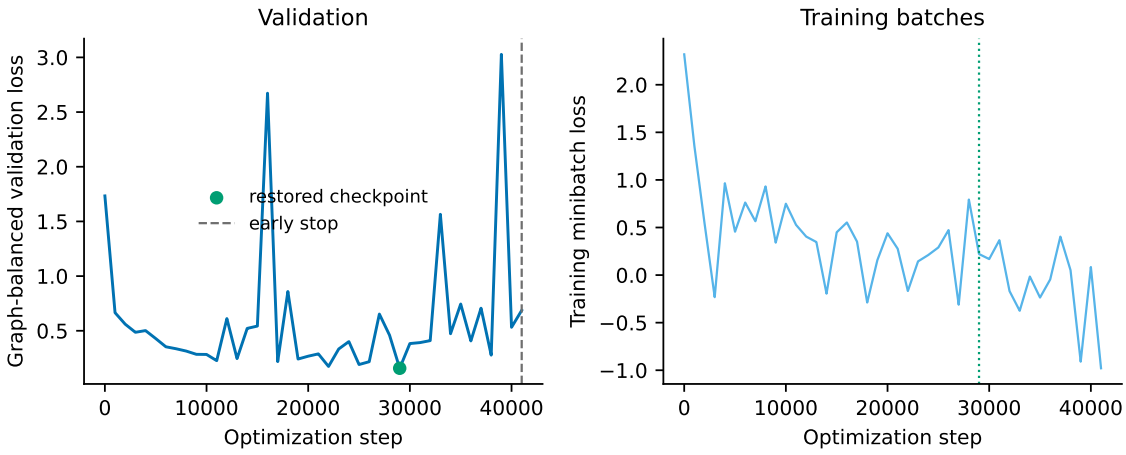


Figure 2: Training of the production model. Validation loss was measured on 1,024 complete graphs at 1,000-step intervals. The green point marks the restored checkpoint and the dashed line marks early stopping. Minibatch loss is noisy because graph size and difficulty differ among batches.

Table 1: Graph-balanced mean performance on 512 independently generated global-landscape examples. Covariance diagnostics were retained for DREEMS only.

Method	Edge corr.	Edge MSE	Cov. corr.	Cov. R2
DREEMS neural	0.698	0.558	0.939	0.917
DREEMS refined	0.720	0.533	0.984	0.976
FEEMS documented LOO-CV	0.614	0.738	–	–

The paired differences were also clear (Table 2). Refined DREEMS increased mean correlation by 0.1058 (95% CI 0.0897–0.1220) and reduced mean MSE by 0.2048 (95% CI 0.1773–0.2323) relative to FEEMS. It had higher correlation in 419 of 512 examples (81.8%) and lower MSE in 429 (83.8%). The per-example comparison shows a broad shift rather than a small number of large gains, although substantial DREEMS failures remain (Figure 3).

The refinement had a larger effect on the quantity it directly targets. Mean off-diagonal covariance correlation increased from 0.939 for the neural surface to 0.984 after refinement, while mean covariance R^2 increased from 0.917 to 0.976. Mean edge correlation increased by 0.0224 and MSE decreased by 0.0243. Covariance fit and edge recovery are not equivalent: sparse observations can be explained by more than one conductance surface.

Accuracy improved as more of the graph was sampled. Refined DREEMS mean correlation was 0.690 in the 430 sparse examples, 0.874 in 73 intermediate examples, and 0.904 in nine dense examples (Table 3). Nevertheless, its group mean exceeded FEEMS in all three strata. In the sparse group, DREEMS increased mean correlation by 0.1146 and reduced mean MSE by 0.2163. The dense stratum is too small for a precise separate comparison.

The average DREEMS advantage was also present in each graph-size stratum (Table 4; Figure 4). The mean correlation advantages were 0.1315 for graphs with 219–450 nodes, 0.0772 for graphs with 451–900 nodes, and 0.0860 for graphs with 901–1,406 nodes. Thus accuracy did not collapse at the largest benchmark size, although all test graphs remained within the training range.

Random examples show that both methods often recovered the largest landscape-scale barriers (Figure 5). DREEMS surfaces were generally smoother and had lower amplitude. In many cases

Table 2: Paired contrasts across the same examples. Positive advantage favors the first method named. MSE advantage is baseline MSE minus candidate MSE.

Paired contrast	N	Mean advantage	95% CI	P	Win frac.
Refined DREEMS vs FEEMS, correlation	512	0.1058	[0.0897, 0.1220]	3.6e-33	0.818
Refined DREEMS vs FEEMS, MSE	512	0.2048	[0.1773, 0.2323]	9.6e-41	0.838
Neural DREEMS vs FEEMS, correlation	512	0.0835	[0.0658, 0.1011]	4.5e-19	0.711
Neural DREEMS vs FEEMS, MSE	512	0.1806	[0.1499, 0.2112]	9.8e-28	0.740
Refinement vs neural, correlation	512	0.0224	[0.0128, 0.0319]	5.4e-06	0.662
Refinement vs neural, MSE	512	0.0243	[0.0140, 0.0345]	4.3e-06	0.607

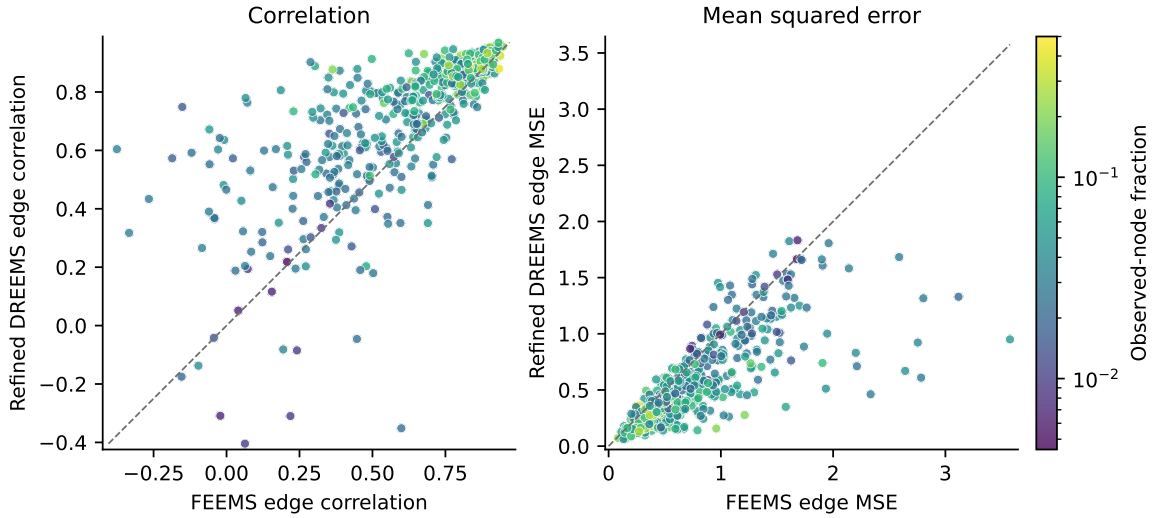


Figure 3: Recovery of simulated edge log-conductance. Each point compares refined DREEMS with documented FEEMS LOO-CV on one landscape. Points above the diagonal in the correlation panel and below it in the MSE panel favor DREEMS. Color gives the fraction of graph nodes containing samples.

only a small fraction of nodes was observed, so estimates across broad unsampled regions were driven largely by resistance structure and the learned simulation prior.

The examples with the largest DREEMS MSE disadvantage show its remaining failure modes (Figure 6). These include misplaced barriers, excessive shrinkage, and incorrect extrapolation through sparsely sampled regions. They are not merely sign-flipped versions of the truth. Because empirical datasets provide no edge truth, such failures cannot be identified by an oracle selector at prediction time.

2.4 DREEMS reduced the cost of repeated analyses

Across 511 process-pool benchmark cases, median complete DREEMS time was 8.4 s and the 95th percentile was 25.8 s. Median FEEMS LOO selection plus final refit was 151.6 s and its 95th percentile was 1,365.6 s (Table 5). The median paired runtime ratio was 16.4 and the geometric mean ratio was 14.8. These timings favor FEEMS computationally: its folds used a process pool of up to 16 workers, whereas DREEMS used two CPU threads. Even so, FEEMS runtime increased sharply with the number of observed demes because each deme creates another cross-validation fold (Figure 7).

Table 3: Accuracy by observed-node fraction. D denotes refined DREEMS and F denotes documented FEEMS LOO-CV.

Sampling	N	D corr.	F corr.	Corr. adv.	D MSE	F MSE	MSE adv.
Sparse ¡=10%	430	0.690	0.575	0.1146	0.583	0.800	0.2163
Intermediate 10-25%	73	0.874	0.811	0.0634	0.273	0.423	0.1507
Dense ¿25%	9	0.904	0.873	0.0316	0.255	0.351	0.0956

Table 4: Accuracy by graph size. Advantages are paired graphwise means.

Nodes	N	D corr.	F corr.	Corr. adv.	D MSE	F MSE	MSE adv.
219-450	252	0.664	0.532	0.1315	0.637	0.870	0.2329
451-900	150	0.765	0.688	0.0772	0.436	0.604	0.1679
901-1,406	110	0.786	0.700	0.0860	0.428	0.619	0.1908

One example with 384 observed demes was run with a separate Snakemake job for every fold. Its ideal compute critical path was 333.9 s, but the folds used 75,904.9 aggregate CPU-seconds before the final refit (Table 6). Parallel execution can reduce latency, but it does not remove the computational work of cross-validation.

2.5 Predictive uncertainty identified difficult graphs

We calibrated the network's predictive standard deviation on independent simulations. On a separate 1,024-graph verification set, graph-balanced coverage was 0.600, 0.857, and 0.957 for nominal 50%, 80%, and 95% intervals (Table 7). The 95% interval was close to nominal for sparse (0.951) and intermediate (0.949) graphs and conservative for the dense group (0.983). The lower-level intervals overcovered overall.

Predictive SD was most informative about broad difficulty (Table 8; Figure 8). Mean SD across a graph had Spearman correlation 0.730 with graph MSE. At the edge level, SD had pooled correlation 0.355 with absolute error. Mean within-graph correlation was 0.203 and was positive in 86.3% of graphs. Edge MSE increased from 0.127 in the lowest-SD quintile to 0.955 in the highest. The uncertainty estimate therefore distinguishes easy from difficult graphs better than it pinpoints every erroneous edge.

The covariance refinement changes the point estimate but does not update the network SD. On the 512-landscape benchmark, centering the calibrated intervals on the refined estimate gave 50%, 80%, and 95% coverage of 0.579, 0.856, and 0.963. These remain predictive intervals learned from simulations, not a posterior distribution conditional on the refinement objective.

2.6 Application to empirical datasets

We applied the same checkpoint without retraining to 15 public datasets (Table 9). They span vertebrates, invertebrates, and plants, with graphs containing 193–4,371 nodes. Fourteen analyses completed in 7.3–16.3 s; the largest sampled-node case, horseweed with 281 observed nodes, took 161.0 s. The resulting surfaces showed broad geographic structure and localized low-conductance regions in several taxa (Figure 9). Because the true migration surfaces are unknown, these analyses establish that the interface and checkpoint operate across diverse public data, not that each inferred feature is correct.

For six examples, we placed the DREEMS result next to a published EEMS, FEEMS, or study figure (Figure 10). Several broad patterns were visually similar, but local structure and amplitude often differed. These panels are qualitative: the original studies can use different variant filters,

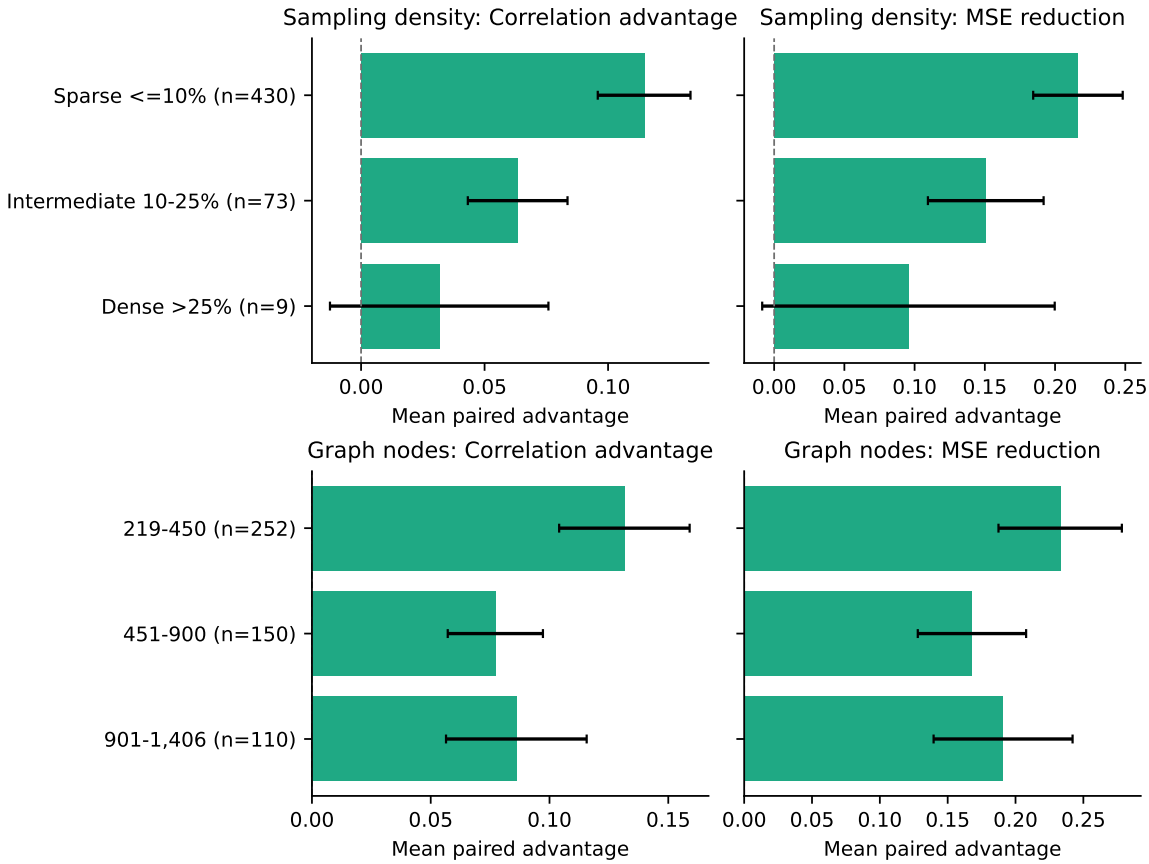


Figure 4: Mean paired DREEMS advantage by sampling and graph-size regime. Positive values favor refined DREEMS for both correlation and MSE reduction. Bars show means and error bars show 95% confidence intervals. The number of graphs differs substantially among strata.

graphs, regularization, models, and rendering. We therefore do not report numerical comparisons with the published images.

2.7 North American gray wolves exposed an empirical limitation

The North American gray-wolf dataset provides a useful empirical test because it was analyzed in the original FEEMS study and is distributed with its software (Schweizer et al., 2016; Marcus et al., 2021). The common-MAF-filtered data contain 111 samples and 17,729 SNPs assigned to 94 of 1,207 graph nodes. We ran DREEMS and the documented 94-fold FEEMS procedure on this same graph and genotype matrix.

The neural DREEMS surface fit the off-diagonal covariance with correlation 0.672 and $R^2 = 0.477$. Refinement increased these values to 0.799 and 0.655. FEEMS reached 0.892 and 0.775 (Table 10). The refined DREEMS and FEEMS edge fields had correlation 0.696, but the DREEMS field had lower edge-log standard deviation (0.784 compared with 1.131).

The difference is visible on the map (Figure 11). FEEMS placed a strong low-conductance region across Hudson Bay and sharper contrasts along coastlines. Refinement moved DREEMS in the same direction relative to the neural estimate, but its mean centered edge log-conductance in the broad Hudson Bay region was -0.074, compared with -0.557 for FEEMS. The graph size is within

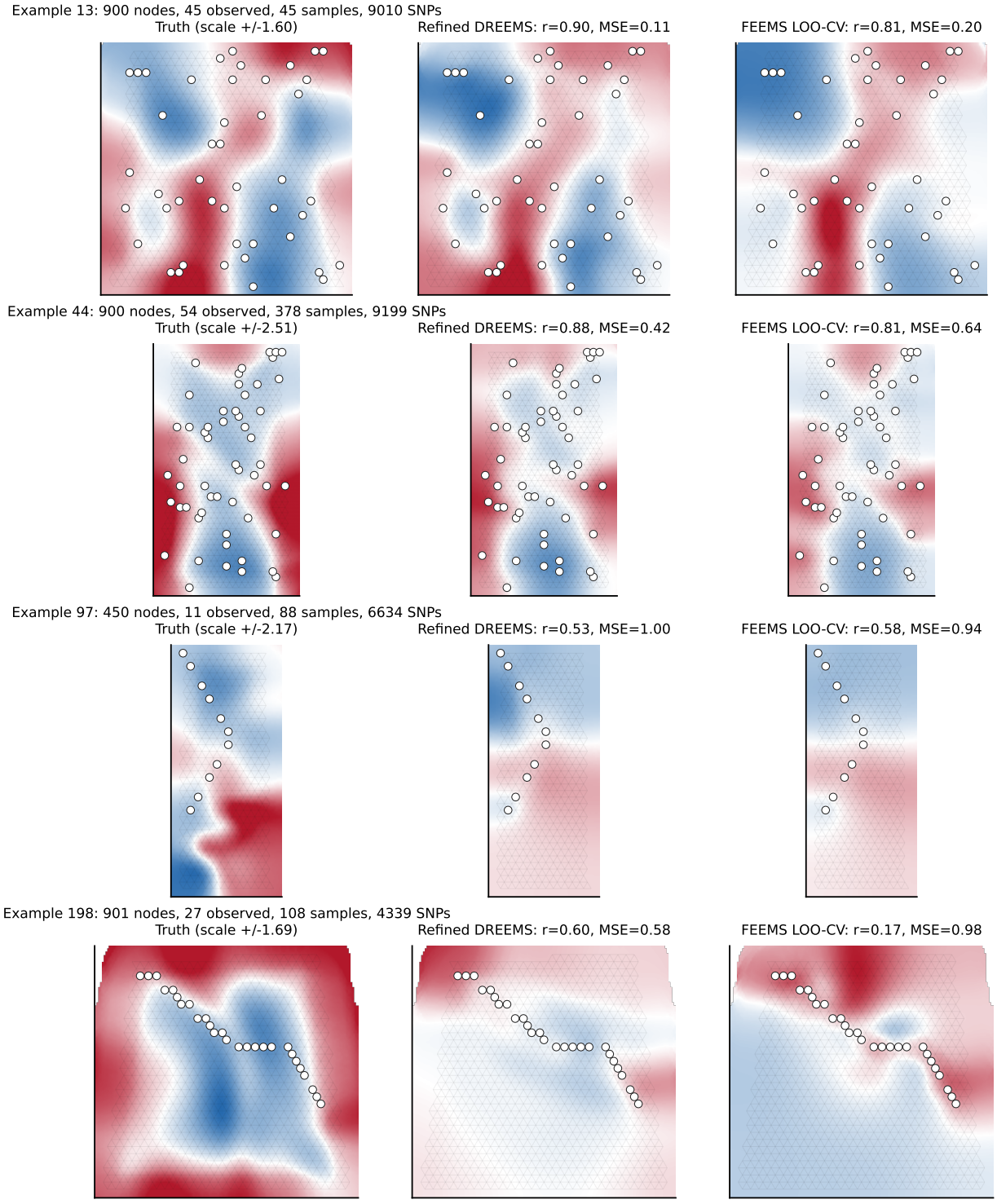


Figure 5: Random examples from the 512-landscape benchmark. Each row shows the true surface, refined DREEMS estimate, and FEEMS estimate on one centered edge-log scale. Circles mark sampled nodes.

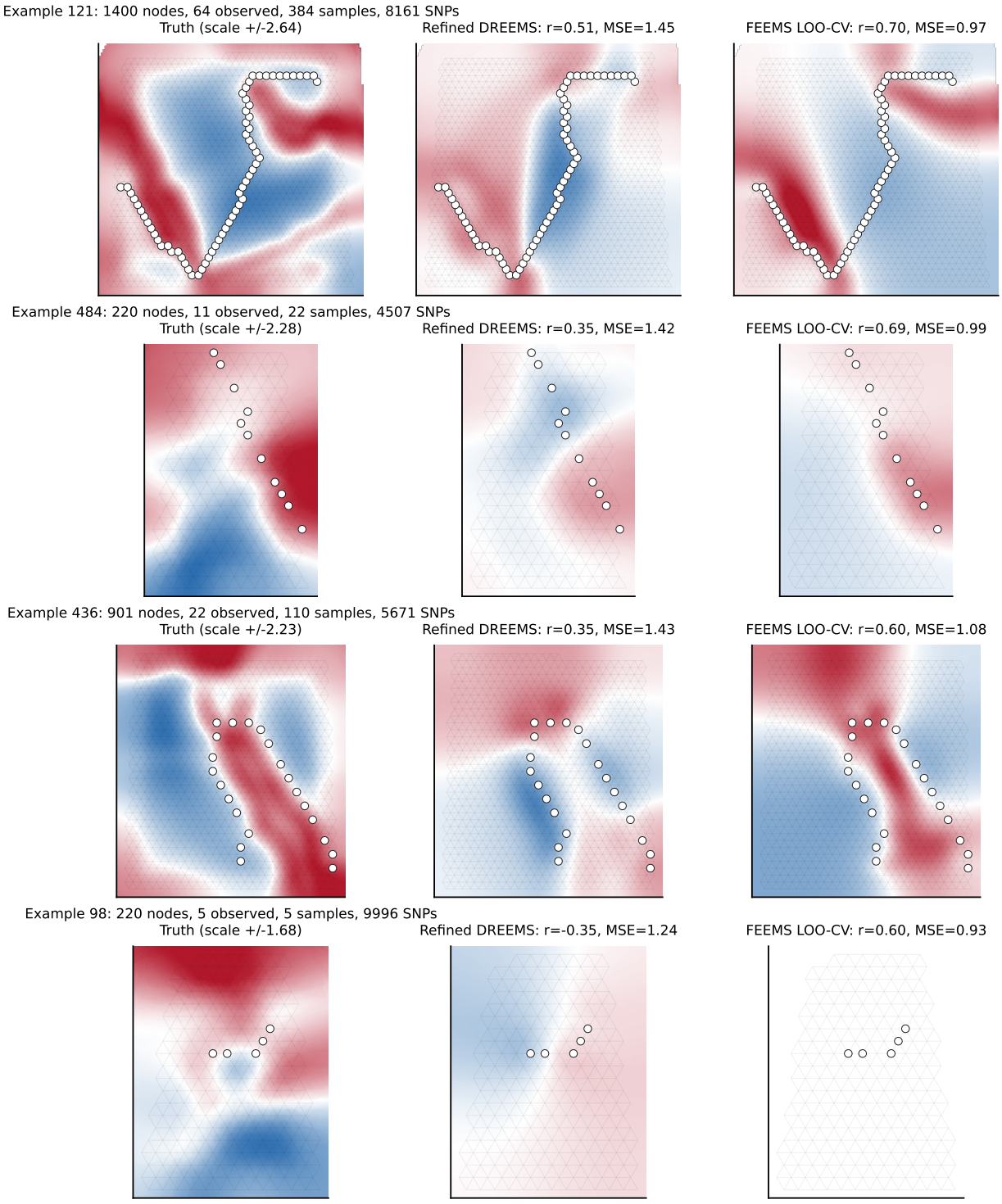


Figure 6: Examples with the largest MSE disadvantage for refined DREEMS. This deliberately unfavorable selection illustrates failure modes and does not summarize average performance.

Table 5: Executed runtime protocols for the 511 process-pool cases. Parallelism was not matched and favors FEEMS.

Execution	N	Parallelism	Median s	P95 s
DREEMS, complete refined estimate	511	2 threads	8.4	25.8
FEEMS LOO-CV and final fit	511	up to 16 workers	151.6	1365.6

Table 6: Resource accounting for the separately atomized 384-fold FEEMS example. Worker-seconds sum wall time over concurrent jobs.

Stage	Jobs	Critical path s	Worker-s	CPU-s
FEEMS LOO folds	384	269.2	83788.4	75904.9
FEEMS selected final fit	1	64.7	64.7	43.1
Combined	–	333.9	83853.1	75948.0

the training range, but only 7.8% of nodes contain samples and the real covariance need not follow the simulated demographic distribution.

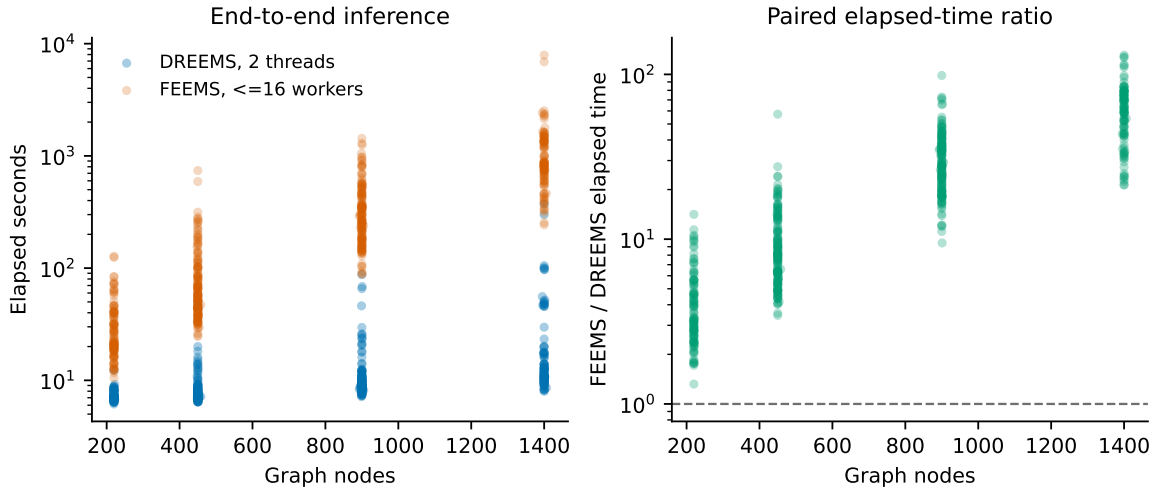


Figure 7: Measured runtime for the process-pool benchmark cases. Complete elapsed time is shown against graph size (left), together with the paired FEEMS-to-DREEMS runtime ratio (right).

Table 7: Marginal predictive-interval coverage. Verification graphs were not used to fit or select the calibrator.

Evaluation	Graphs	50%	80%	95%
Independent verification: all	1024	0.600	0.857	0.957
Independent verification: sparse ¡=10%	610	0.554	0.833	0.951
Independent verification: medium 10-25%	211	0.590	0.842	0.949
Independent verification: dense ¿25%	203	0.750	0.944	0.983
Landscape H2H: neural	512	0.564	0.846	0.958
Landscape H2H: refined mean, neural SD	512	0.579	0.856	0.963

Table 8: Association between calibrated predictive SD and refined edge error on 1,056,285 edges from the 512 global-landscape examples.

Diagnostic	Value
Pooled SD vs absolute refined error, Spearman	0.355
Mean within-graph Spearman	0.203
Graphs with positive within-graph association	0.863
Mean SD vs graph MSE, Spearman	0.730
Lowest-SD / highest-SD quintile MSE	0.127 / 0.955

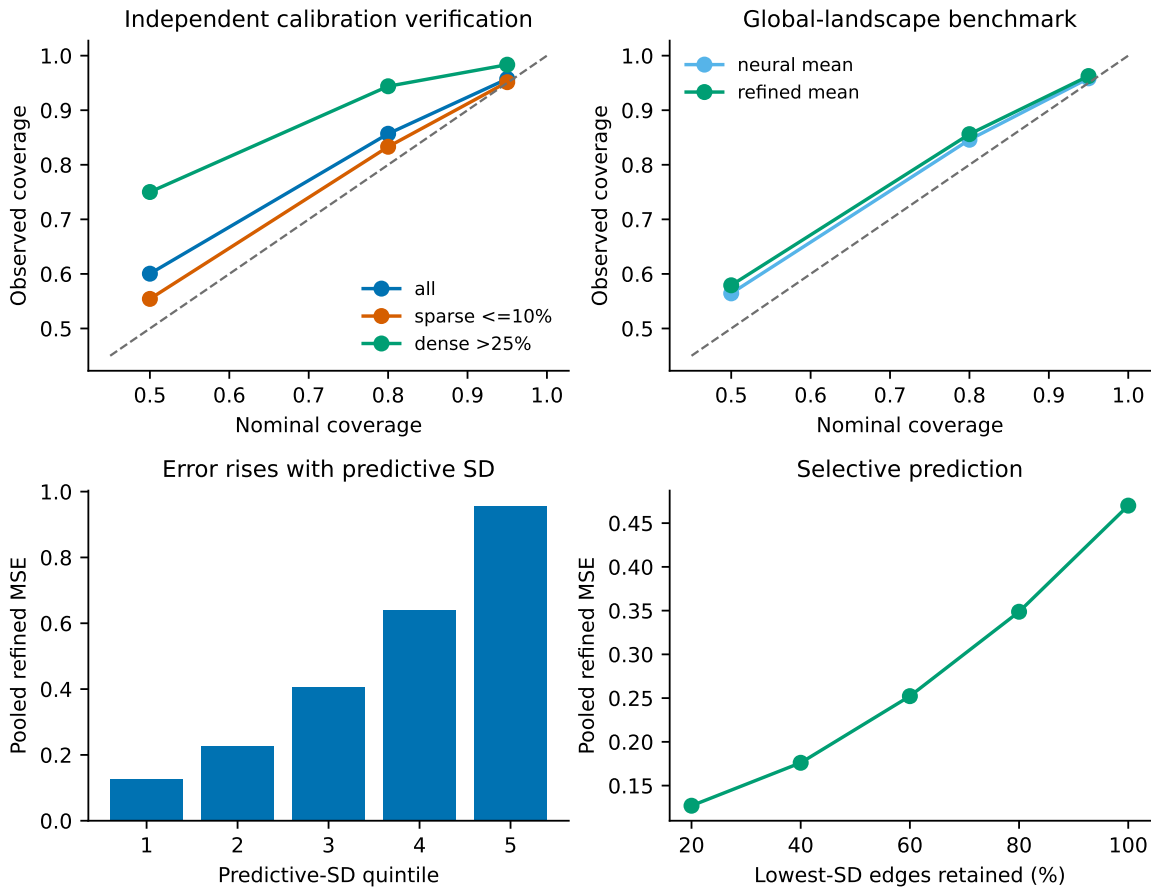


Figure 8: Calibration and utility of predictive SD. Coverage is averaged by graph. Error summaries use the refined mean and the unchanged calibrated neural SD.

Table 9: Public empirical datasets analyzed with one checkpoint. Obs. is the number of graph nodes containing samples.

Dataset	Samples	SNPs	Nodes	Obs.	Edges	Seconds
Gray wolves	111	17729	1207	94	3458	16.3
I. scapularis	517	9197	511	42	1463	8.9
C. cearae	61	13681	418	16	1183	8.5
S. cearensis	48	13423	359	16	1023	7.6
P. mcallii	42	3731	262	16	704	8.0
Poplar	434	20601	773	35	2232	8.4
Allohistium	97	3333	516	39	1444	9.4
C. galerita	88	5990	557	38	1491	7.9
B. tryoni	293	6540	193	27	539	7.3
P. cinereus	38	232	1192	38	3394	9.8
H. hajarensis	30	1113	998	18	2901	9.7
H. luqueorum	23	1170	999	10	2905	8.0
R. forreri	103	172	4371	29	12829	15.7
E. canadensis	639	5080	1374	281	4029	161.0
S. cayana	83	19031	791	49	2310	9.0

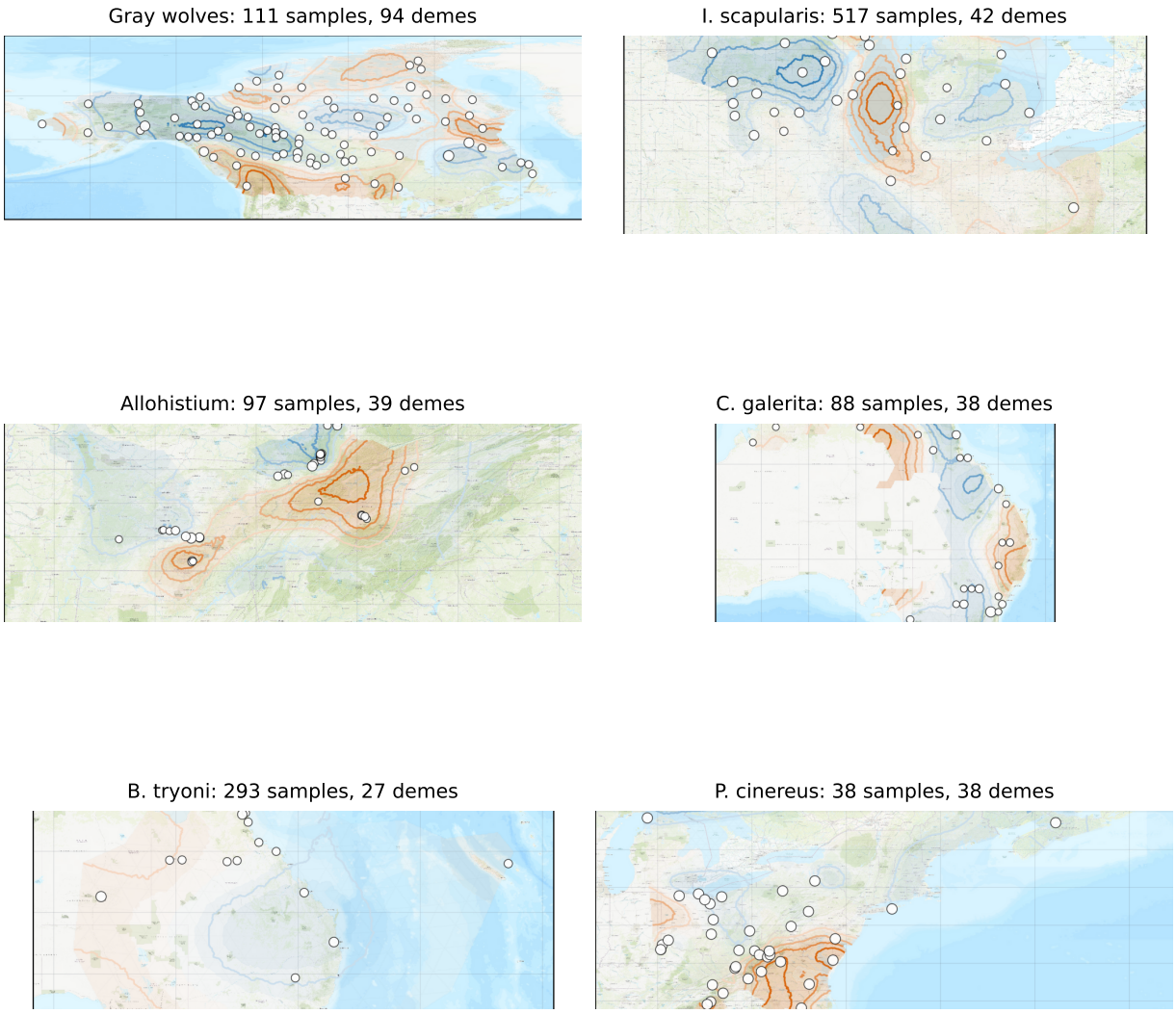


Figure 9: Selected empirical DREEMS surfaces. Blue indicates higher and red lower centered relative migration; white is near the graph mean. Maps and contours interpolate edge estimates for display only.

Table 10: North American gray-wolf diagnostics. Edge correlation is relative to the FEEMS field, not to unknown truth. Hudson mean is a descriptive centered edge-log average over a fixed bounding region.

Method	Cov. corr.	Cov. R2	Cov. MSE	Edge corr. vs F	Edge SD	Hudson mean
DREEMS neural	0.672	0.477	0.001138	0.409	0.664	0.040
DREEMS refined	0.799	0.655	0.000751	0.696	0.784	-0.074
FEEMS 94-fold LOO-CV	0.892	0.775	0.000489	1.000	1.131	-0.557

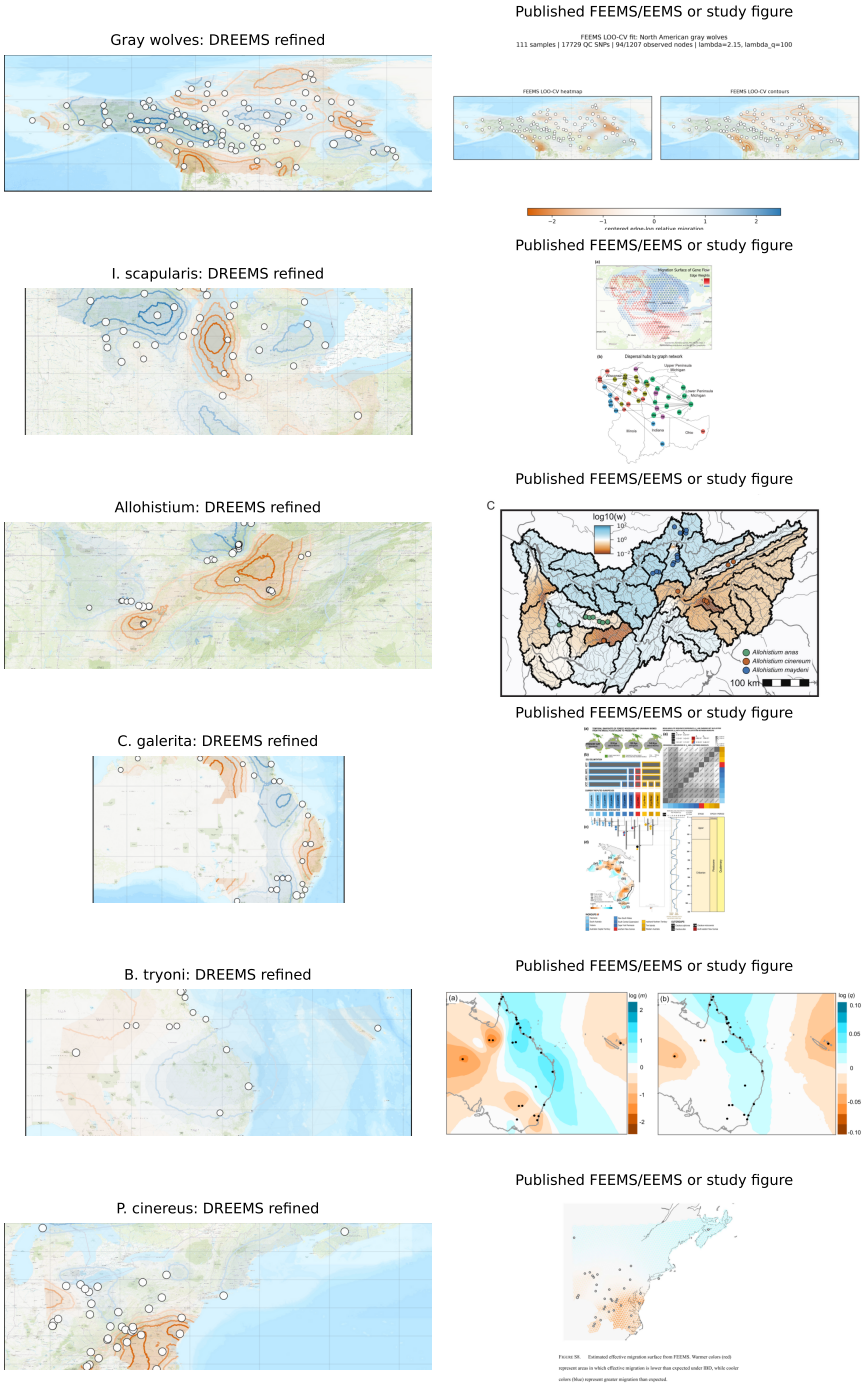


Figure 10: Qualitative comparison with published FEEMS/EEMS or study figures. Rows need not share graph, filtering, model, tuning, scale, or rendering, so no numerical comparison is implied.

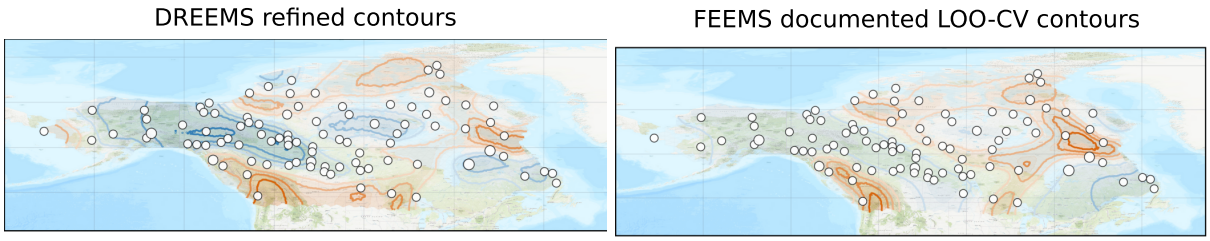


Figure 11: North American gray-wolf surfaces using the packaged graph and the same common-MAF-filtered data. Refined DREEMS is shown at left and FEEMS, selected by documented joint LOO cross-validation, at right.

3 Discussion

DREEMS provides a fast way to estimate effective migration surfaces from genetic covariance. On 512 simulated landscapes, the refined estimate was more accurate on average than FEEMS run with joint leave-one-deme-out selection of migration and local-drift penalties. The advantage persisted in the sparsely sampled group and across the tested range of graph sizes. Once trained, one checkpoint could also be applied to 15 public datasets without a new hyperparameter search.

The comparison measures two different ways of regularizing a difficult inverse problem. FEEMS chooses a smooth surface separately for each dataset. DREEMS learns regularities across many simulated graphs and reuses them. More than half of the training corpus was generated by the same global-landscape meta-procedure as the synthetic benchmark, although every benchmark graph, surface, sampling design, and random seed was new. The simulation result is therefore evidence for amortization over this family, not a guarantee under an arbitrary demographic model.

Several components helped make the learned estimate useful across graphs. The linearized resistance calculation gives every observed pair a way to inform edges that carry current between them. Diffusion bands retain genetic and sampling patterns at several graph scales instead of relying only on repeated local message passing. Graph-wide attention supplies long-range context. The representation is also invariant to translation, rotation, reflection, uniform scale, and node relabeling. This last property prevents the drawing of a graph from creating a preferred migration direction.

The fixed covariance refinement combines the speed of a neural prediction with a small amount of dataset-specific correction. It improved covariance fit substantially and edge recovery modestly. The trust region is important because many edge surfaces can explain a sparse observed covariance. Optimizing the forward fit without this restriction can move toward a statistically similar but spatially incorrect surface. Training through the refinement is possible in principle, but would require a separate experiment to show that its added cost improves generalization.

Predictive uncertainty supplies another diagnostic. After calibration, 95% marginal intervals had nearly nominal coverage on independent simulations, and mean SD distinguished difficult from easy graphs. Local edge ranking was weaker, and the factorized predictor does not describe dependence among edge errors. The intervals also retain the neural SD after covariance refinement. They should be read as simulation-based warning measures rather than Bayesian credible intervals for an empirical dataset.

The wolf analysis illustrates the central limitation. DREEMS was competitive on the simulated landscape benchmark, yet FEEMS fit the empirical wolf covariance better and recovered stronger structure around Hudson Bay. DREEMS receives no water or elevation layer, so it can recover a shoreline only when genetic covariance, sample placement, graph geometry, and its learned prior support that interpretation. The weaker wolf result suggests that this combination of sparse sampling, geographic scale, and demographic history is not fully represented by the training distribution.

Other limitations follow from the model itself. Relative effective migration is not an identified contemporary migration rate. The present checkpoint assumes a connected undirected graph, diploid simulations, and covariance from common variants. It has not been validated for directional migration, strong linkage, rare-variant summaries, polyploid genotypes, or graphs substantially larger than those used in training. Interpolated maps can also appear more continuous and geographically precise than the underlying edge estimates.

We therefore view DREEMS as a rapid first analysis rather than a substitute for scrutiny of the data and graph. Its clearest use is exploratory work across many datasets or sensitivity analyses that would make repeated cross-validation costly. For strong empirical claims, users should examine covariance fit, predictive uncertainty, graph construction, and variant filtering, and compare with a dataset-specific fit when the inferred surface is unstable or biologically implausible.

4 Materials and methods

4.1 Model description

Migration as graph conductance. We represent a habitat by a connected, undirected graph $G = (V, E)$ with $n = |V|$ nodes and $m = |E|$ edges. Let $B \in \mathbb{R}^{m \times n}$ be an incidence matrix under an arbitrary edge orientation. The target is a vector of relative edge log-conductances $\boldsymbol{\theta} \in \mathbb{R}^m$. Positive conductances $w_e = \exp(\theta_e)$ define the weighted graph Laplacian

$$L(\boldsymbol{\theta}) = B^{\mathsf{T}} \operatorname{diag}\{\exp(\boldsymbol{\theta})\} B. \tag{1}$$

The effective resistance between nodes i and j is

$$R_{ij}(\boldsymbol{\theta}) = (\boldsymbol{e}_i - \boldsymbol{e}_j)^{\mathsf{T}} L(\boldsymbol{\theta})^{+} (\boldsymbol{e}_i - \boldsymbol{e}_j), \tag{2}$$

where L^+ is the inverse on the mean-zero subspace. Multiplying every conductance by the same constant changes the covariance scale but not the relative surface. We therefore center true and estimated log-conductances to have mean zero within each graph.

The canonical input is a node-level covariance matrix, coordinates for the graph nodes, an undirected edge list, indices and sample counts for observed nodes, and an effective SNP count. The public interface also accepts a sample covariance matrix and sample coordinates, or genotypes and sample coordinates, and constructs this object. Geographic coordinates assign samples to the nearest node and determine normalized edge length. Environmental layers and a fitted FEEMS surface are not inputs.

The model also predicts a centered node field $\eta_i = \log q_i - \operatorname{mean}_j \log q_j$, where q_i is the local coalescence-rate coefficient used by the simulator. This nuisance output helps the network distinguish variation in within-deme drift from edge conductance, although the two remain statistically coupled.

Covariance calculation and deme aggregation. For a diploid genotype matrix X, with s samples, ℓ SNPs, and entries encoded as 0, 1, or 2, we remove monomorphic variants and standardize each retained SNP across samples:

$$Z_{al} = \frac{X_{al} - \bar{X}_{\cdot l}}{\operatorname{sd}(X_{\cdot l})}, \qquad C = \ell^{-1} Z Z^{\mathsf{T}}. \tag{3}$$

The sample standard deviation uses denominator $s - 1$. Missing genotypes in the empirical analyses were mean-imputed after variant filtering. If A averages samples assigned to each observed node, the node-level covariance is $C_o = ACA^{\mathsf{T}}$.

When a user supplies covariance directly, we symmetrize it and project small negative eigenvalues to zero before aggregation. The effective SNP count is retained because it controls sampling-precision features. Covariance input does not preserve linkage, allele-frequency spectra, or higher-order genotype information.

Covariance score maps. We retain the leading 20 eigenpairs of the observed-node covariance and form score maps

$$U_k = \sqrt{\lambda_k}\, \boldsymbol{v}_k, \qquad k = 1, \ldots, 20. \tag{4}$$

Scores at unobserved nodes are zero, eigenvector signs are fixed deterministically, and unavailable components are zero-padded. These are eigenvectors of genetic covariance, not graph-Laplacian eigenvectors.

4.2 All-pairs linearized resistance initializer

The o observed demes provide $o(o-1)/2$ pairwise dissimilarities

$$D_{ij} = C_{o,ii} + C_{o,jj} - 2C_{o,ij}. \tag{5}$$

The initializer uses every observed pair. Its implementation batch size of 256 limits working memory and does not subsample the data.

We define a fixed baseline conductance $w_e^{(0)} = \mathrm{clip}(d_{\mathrm{med}}/d_e, 0.25, 4)$ from edge length. Sparse grounded solves give baseline resistances $R_{ij}^{(0)}$ and node potentials $\boldsymbol{\phi}_{ij}$ for unit current injected at i and removed at j. The first-order sensitivity of log resistance to edge log-conductance is

$$A_{(ij),e} = \left.\frac{\partial \log R_{ij}}{\partial \theta_e}\right|_{\boldsymbol{\theta}^{(0)}} = -\frac{w_e^{(0)}\{(B\boldsymbol{\phi}_{ij})_e\}^2}{R_{ij}^{(0)}}. \tag{6}$$

Thus a pair contributes to each edge carrying baseline electrical flow between its endpoints.

Let $\boldsymbol{r}$ be the centered difference between observed and baseline log dissimilarity. We accumulate $\boldsymbol{g} = A^{\mathsf{T}}\boldsymbol{r}$ and the diagonal information approximation $h_e = \sum_{i<j} A_{(ij),e}^2$ without materializing the dense sensitivity matrix. The update solves

$$[\mathrm{diag}(\boldsymbol{h} + \rho) + 0.01 L_E]\,\boldsymbol{\delta} = \boldsymbol{g}, \tag{7}$$

where L_E is the unweighted line-graph Laplacian and ρ is 0.001 times the median positive diagonal information. We center and scale $\boldsymbol{\delta}$ to unit standard deviation. The field and a numerical-validity indicator are the final two edge features. This is a one-step diagonal approximation. It is evidence for the network to denoise rather than a stand-alone inverse solution.

4.3 Fixed diffusion-band features

The spatial representation is unchanged by translating, rotating, reflecting, uniformly rescaling, or relabeling the same graph. The network receives no absolute coordinates or signed x/y edge displacement. Coordinate slots retained for checkpoint-shape compatibility are fixed at zero. Edge length is divided by a graph-level reference length, endpoint sample-count and precision features use symmetric means and absolute differences, and support is routed over all shortest paths rather than one arbitrarily selected path.

For the multiscale features, the reference edge weight is $a_e = \mathrm{clip}(d_{\mathrm{med}}/d_e, 0.25, 4)$. If A_0 and D_0 are the corresponding adjacency and degree matrices, the fixed normalized Laplacian is

$$L_0 = I - D_0^{-1/2} A_0 D_0^{-1/2}. \tag{8}$$

This operator depends on graph topology and relative edge length, not on the unknown migration surface. For a node-signal matrix S, we compute $S_t = \exp(-tL_0)S$ at $t = (0.25, 0.5, 1, 2, 4)$ and form

$$\begin{aligned} \Psi_0(S) &= S - S_{0.25}, \\ \Psi_k(S) &= S_{t_k} - S_{t_{k+1}}, \quad k = 1, \ldots, 4, \\ \Psi_5(S) &= S_4. \end{aligned} \tag{9}$$

The bands retain changes removed at successive diffusion scales and a final coarse component.

The diffusion input contains 13 standardized node signals: the first eight covariance score maps, log sample count, a sampled-node indicator, total and within-deme covariance support, and the

covariance diagonal. Six bands give 78 node features. For each signal and band, we compute three symmetric edge summaries: absolute endpoint contrast, log one plus squared contrast, and mean endpoint absolute energy. These give 234 edge features. Base features add the full 20 covariance scores, sample and SNP precision, covariance support, normalized length, symmetric endpoint summaries, and shortest-path support. The final arrays have 109 node and 253 edge columns, including the two initializer columns. Training-corpus means and standard deviations normalize each column, with a standard-deviation floor of 0.2.

4.4 Graph neural network

Table 11 summarizes the model. We implemented it with Equinox and JAX (Bradbury et al., 2018; DeepMind et al., 2020; Kidger and Garcia, 2021). An affine GELU layer embeds node features in 128 dimensions. Ten residual edge-aware message-passing layers apply directed messages

$$\boldsymbol{m}_{j\to i}^{(k)} = \boldsymbol{h}_j^{(k)} \odot \mathrm{sigmoid}(W_g^{(k)}\boldsymbol{x}_{ij}) + \mathrm{GELU}(W_m^{(k)}\boldsymbol{x}_{ij}), \tag{10}$$

followed by a residual node update using the mean incoming message. A three-layer context branch uses the same graph. No graph coarsening or pooling is present in the released architecture.

For edge (i, j), the decoder uses

$$\boldsymbol{u}_{ij} = [\boldsymbol{h}_i + \boldsymbol{h}_j, |\boldsymbol{h}_i - \boldsymbol{h}_j|, \boldsymbol{x}_{ij}], \tag{11}$$

which is symmetric in its endpoints. A query derived from $\boldsymbol{u}_{ij}$ attends to keys and values from all context-node embeddings, giving every edge graph-wide context despite finite message-passing depth. Separate affine heads predict an edge mean μ_e and log variance ℓ_e, with $\ell_e \in [-7, 5]$, and analogous node-q outputs. Edge means and node means are centered within each graph.

The output family is a factorized normal distribution trained against simulated targets. It approximates the conditional distribution induced by the simulation corpus. It is not the result of variational inference for an explicit per-dataset likelihood, and it does not represent covariance among edge errors. We therefore call $\exp(\ell_e/2)$ the network predictive SD, not a posterior SD.

4.5 Graph-balanced training objective

Let $\theta_e^\star$ and $\eta_i^\star$ be centered simulation targets. The principal edge term is the graphwise heteroscedastic Gaussian negative log-likelihood

$$\mathcal{L}_{\mathrm{edge}} = \frac{1}{2m}\sum_{e=1}^{m}\left[(\mu_e - \theta_e^\star)^2 \exp(-\ell_e) + \ell_e\right]. \tag{12}$$

The node-q head uses the same form. The full loss is

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\mathrm{edge}} &+ 0.5\mathcal{L}_{\mathrm{context}} + 0.5\mathcal{L}_q \\ &+ 0.25\mathcal{L}_{\mathrm{slope}} + 0.20\mathcal{L}_{\mathrm{corr}} + 0.35\mathcal{L}_{\mathrm{orientation}}.\end{aligned} \tag{13}$$

The context term is edge MSE from the auxiliary branch. The slope term is a Huber loss on the within-graph regression slope relative to one, the correlation term is $1 - r$, and the orientation term penalizes failure to separate above- and below-mean target edges. Shape terms are disabled when the target's centered RMS is at most 0.05. Every component is reduced within each graph before averaging over the minibatch, so a graph contributes one weight regardless of its edge count. The training loss contains no resistance or covariance-forward term.

Table 11: Model and optimization settings for the production checkpoint.

Component	Setting
Node / edge dimensions	109 / 253
Hidden channels	128
Local / context graph layers	10 / 3
Graph-wide attention	yes
Covariance score maps	20
Diffusion times	0.25, 0.5, 1, 2, 4
Spatial representation	topology and normalized edge length
Coordinate-frame invariant	translation, rotation, reflection, scale
Linearized initializer	all_pair_diagonal_adjoint
Initializer pairs	all observed-deme pairs
Predictive family	factorized_normal
Training examples	61,440
Validation / held-out examples	1,024 / 4,096
Best / final step	29,000 / 41,000
Best validation loss	0.1572
Optimizer / learning rate / batch	Adam / 5e-5 / 8

4.6 Synthetic training corpus

Graphs, surfaces, and sampling. We generated 61,440 training examples, 1,024 validation examples, and 4,096 held-out examples with disjoint random seeds. Table 12 reports the realized training ranges. The corpus combines five graph families.

The largest component contains 34,605 random global-landscape examples. We sample geographic windows over land between 55°S and 72°N from Natural Earth 50-m polygons (Natural Earth, 2026), project each window to a local azimuthal-equidistant system, and place a hexagonal mesh over its full bounding box. Water nodes remain in the graph but cannot be sampled. Target sizes are 220, 450, 900, or 1,400 nodes. Conductance surfaces combine a random smooth field, habitat mismatch, edge length, water and coast penalties, and a randomly placed sinuous ridge. Coefficients, length scales, and final edge-log standard deviation vary among examples. A separate smooth field generates local q. Natural Earth and latent habitat variables generate targets and sampling designs; they are never supplied to DREEMS.

Another 12,256 examples use the 8×12 triangular lattice from FEEMS simulations, with homogeneous, barrier, and anisotropic surfaces and either full or approximately 20% node sampling. A further 5,123 examples use land or synthetic-corridor graphs with wolf-like sampling paths. The remaining 9,456 use masked grids and earth-like graphs with smooth, barrier, Gaussian-process, or constant surfaces. Across global-landscape and wolf-like examples, requested sampled-node fractions are 0.03, 0.05, 0.10, 0.25, or 0.50, weighted toward the sparse values. Each sampled node contains one to eight diploid individuals, up to 384 total in those families. Other graph families extend the realized range to 1,200 samples.

Structured-coalescent SNP simulation. We wrote a Rust simulator for large stepping-stone models. Its state consists of ancestral lineages and their current demes. If deme i contains k_i lineages, coalescence occurs there at total rate $q_i k_i(k_i - 1)/2$; each lineage moves from i to j at the specified edge migration rate. Production rates are symmetric. Descendant sets are stored in one or more 128-bit words, and random draws use Rust `SmallRng`.

For the production `conditioned_random_marker` scheme, we simulate a genealogy and select a

Table 12: Realized production training corpus. Counts refer to graphs, not SNPs.

Corpus property	Value
Total	61,440
Global landmass graphs	34,605
FEEMS triangular lattices	12,256
Wolf-like graphs	5,123
Masked grids and earth-like graphs	9,456
Nodes	9-1,743 (median 450)
Edges	12-4,107 (median 1,255)
Observed-node fraction	0.002-1.000 (median 0.091)
Diploid samples	2-1,200 (median 200)
Accepted SNPs	2,000-10,000 (median 5,994)
Edge log-rate SD	0.350-1.600 (median 0.976)
Node log-q SD	0.000-0.900 (median 0.452)
Simulator	Rust structured coalescent
Ascertainment	random marker conditional on MAF ¿=0.05

branch proportional to branch length, which is the location distribution of a random infinite-sites mutation conditional on a mutation occurring on that genealogy. We retain a marker only when its sample allele frequency lies in $[0.05, 0.95]$ and repeat until reaching the requested number. Within demes, derived and ancestral haplotypes are randomly paired without replacement to form diploid dosages. Accepted dosage columns are standardized and accumulated directly into Equation 3; genotype matrices are not stored for the training corpus. Examples contain 2,000–10,000 accepted SNPs.

We compared the Rust simulator with `msprime` on small graphs and with symmetry and closed-form checks for the symmetric island model (Baumdicker et al., 2022). Tests compared allele counts, pairwise differences, and covariance summaries over replicates.

4.7 Optimization and checkpoint selection

We used Adam with learning rate 5×10^{-5}, minibatch size 8, and global gradient-norm clipping at one (Kingma and Ba, 2015). Training sampled graphs repeatedly rather than making one pass through the corpus. We evaluated all 1,024 validation graphs every 1,000 steps. After a 20,000-step warmup, early stopping required an improvement greater than 10^{-4} within 12 validation checks. The best graph-balanced validation loss was 0.15724 at step 29,000. Training stopped at step 41,000 and restored the step-29,000 parameters. The one-GPU run took 7,578.6 seconds (2.11 h), used 2,230.1 CPU-seconds, and reached 5.35 GiB resident memory. The checkpoint SHA-256 is `f0b677d2cc028252002d15fac843ca7347b1b352eb2286a18bd202b724aa92b4`.

4.8 Fixed q-aware trust-region covariance refinement

The neural prediction is amortized and need not closely fit one dataset's covariance. The default pipeline therefore applies a fixed 50-step refinement without selecting a test-specific hyperparameter. Let $P_o = I_o - \mathbf{1}\mathbf{1}^\mathsf{T}/o$ and let $G_o(\boldsymbol{\theta})$ be the observed-node submatrix of $L(\boldsymbol{\theta})^+$. Define

$$\bar{C}_o = P_o C_o P_o, \qquad \bar{G}_o(\boldsymbol{\theta}) = P_o G_o(\boldsymbol{\theta}) P_o. \tag{14}$$

From the predicted node field we also form

$$Q_o(\boldsymbol{\eta}) = \operatorname{diag}\left\{ \frac{\exp(\eta_i)}{n_i} : i \text{ observed} \right\}, \qquad \bar{Q}_o = P_o Q_o P_o, \tag{15}$$

where n_i is the number of samples at observed node i.

At every edge update, nonnegative least squares fits global scales $\alpha, \beta \geq 0$ for the migration and local-q bases. The objective is

$$\begin{aligned} \mathcal{J}(\boldsymbol{\theta}) = \frac{1}{2\binom{o}{2}} \sum_{i<j} \left\{ \bar{C}_{o,ij} - \alpha \bar{G}_{o,ij}(\boldsymbol{\theta}) - \beta \bar{Q}_{o,ij} \right\}^2 \\ + \frac{10^{-4}}{2m} \boldsymbol{\theta}^\top L_E \boldsymbol{\theta}. \end{aligned} \tag{16}$$

Analytic gradients through sparse grounded solves avoid constructing L^+. We use 50 Adam updates at learning rate 0.15. Edge batches of 256 control working memory only.

The update remains in an uncertainty-scaled trust region around the neural mean $\boldsymbol{\mu}$:

$$\left[\frac{1}{m} \sum_{e=1}^{m} \left\{ \frac{\theta_e - \mu_e}{\max(\sigma_e^{\text{raw}}, 0.25)} \right\}^2 \right]^{1/2} \leq 0.5. \tag{17}$$

The radius uses the raw neural SD because it was fixed before post-hoc calibration. The locked configuration has no Gaussian-prior penalty; its weight is zero. We therefore call the procedure constrained covariance refinement rather than MAP. Refinement changes the point estimate but not the reported predictive SD.

4.9 Synthetic head-to-head benchmark

Test examples. We generated 512 independent global-landscape examples with new random seeds. Graphs contain 219–1,406 nodes (median 452) and 590–4,065 edges (median 1,270). They contain 2–384 observed nodes (median 39), 5–384 diploid samples (median 135), and 2,003–9,996 accepted SNPs (median 5,811). The observed-node fraction ranges from 0.004 to 0.500 (median 0.049). We evaluate both the frozen neural mean and the fixed radius-0.5 refined mean.

Documented FEEMS LOO cross-validation. We pinned upstream FEEMS 2.0 at commit `aa890b0e8ef583dd657a6a696af591b732d1d8be`. Following its documented cross-validation notebook and source, each fold holds out all samples at one observed deme. Candidate fits use the remaining demes, and selection scores held-out allele-frequency prediction by L2 error. We jointly search ten migration-smoothness values logarithmically spaced from 100 to 0.001 and five q-smoothness values from 100 to 0.01. The selected pair is refit to all samples with node-specific q, unbounded log weights, and at most 15,000 iterations. This gives FEEMS its documented leave-one-observed-deme-out, dataset-specific selector; DREEMS uses one checkpoint and one prespecified refinement for every example. Both methods receive the same genotype matrix, sample locations, and graph.

Accuracy and paired inference. For each graph, we center estimates and true edge log-conductances and compute Pearson correlation and mean squared error (MSE). Each graph contributes one value, regardless of edge count. For paired contrasts, an advantage is DREEMS correlation minus FEEMS correlation or FEEMS MSE minus DREEMS MSE. We report the mean

advantage, a two-sided 95% Student-t interval, the paired t-test P value, and the fraction of examples won. Sampling strata are sparse ($\leq 10\%$ of graph nodes observed), intermediate ($> 10\%$ and $\leq 25\%$), and dense ($> 25\%$). Graph-size strata were fixed before summarization.

For DREEMS only, we also evaluate covariance fit by correlating observed off-diagonal centered covariance with $\alpha\bar{G}_o + \beta\bar{Q}_o$ and by its R^2 and MSE. Comparable covariance predictions were not retained in the completed synthetic FEEMS artifacts, so these values are diagnostic rather than a cross-method covariance comparison.

4.10 Runtime and resource accounting

Runtime includes model loading, feature construction, the neural forward pass, and refinement for DREEMS, and all LOO folds plus the selected final refit for FEEMS. In 511 cases, DREEMS used two CPU threads and FEEMS executed folds through a process pool of up to 16 workers. The remaining example had 384 FEEMS folds, each represented as an independent Snakemake job (Köster and Rahmann, 2012); we report its idealized compute critical path and aggregate worker use separately. These measurements describe the executed workflows, not hardware-normalized single-core benchmarks. Training and corpus-generation costs are not charged to an individual DREEMS analysis.

4.11 Predictive-SD calibration and utility

We used three independent sets of 1,024 simulated graphs for calibration fit, model selection, and verification. Candidate calibrators left raw SDs unchanged, multiplied all SDs by one global scale, or applied additional sampling-density factors for fractions ≤ 0.10, 0.10–0.25, and > 0.25. Parameters were fit by graph-balanced normal NLL, and the candidate with lowest NLL on the selection set was locked. The selected density calibrator had global multiplier 1.3117 and density factors 1.0753, 0.9951, and 1.9091. We then measured graph-balanced 50%, 80%, and 95% marginal coverage on the verification set. The same locked calibration was used in the 512-example head-to-head benchmark.

To ask whether uncertainty is useful beyond coverage, we correlated calibrated edge SD with absolute refined error, computed within-graph rank correlations, correlated graph-mean SD with graph MSE, and compared MSE across pooled SD quintiles. These are marginal diagnostics for a factorized predictor, not simultaneous coverage statements.

4.12 Empirical reanalyses

We applied the checkpoint without retraining to 15 public datasets that had georeferenced genotypes, covariance-like EEMS input, or an existing EEMS/FEEMS analysis. For raw genotypes we retained biallelic variants, required study-appropriate call rate, mean-imputed missing calls, and applied a common minor-allele-frequency threshold of 0.05. Exact source, filtering, graph construction, and SNP-count provenance is stored per dataset in the release manifest. Graphs contain 193–4,371 nodes and 539–12,829 edges; observed fractions range from 0.0066 to 0.2045.

Published comparison panels were copied from the original papers or author-distributed EEMS output. They can differ from our reanalysis in input filtering, graph construction, regularization, method, and rendering. We use them only for a qualitative visual audit. Because true conductance fields are unknown in empirical data, no edgewise accuracy is reported.

The North American gray-wolf data distributed with FEEMS received a separate same-graph analysis (Schweizer et al., 2016; Marcus et al., 2021). The common-MAF filtered matrix contains 111 samples and 17,729 SNPs assigned to 94 of 1,207 nodes on the packaged 3,458-edge graph. We compare neural and refined DREEMS with FEEMS selected by its 94-fold joint LOO procedure. As

descriptive diagnostics, we report off-diagonal covariance fit, correlation of each DREEMS edge field with the FEEMS field, edge-field standard deviation, and mean centered edge log-conductance in a broad Hudson Bay bounding region. The last quantity is not a formal geographic hypothesis test.

Data and code availability

The software includes the Equinox checkpoint, an xarray-based Python API, command-line entry points, the Rust simulator, tests, and Snakemake workflows for corpus generation, training, calibration, FEEMS comparison, and figure construction. The exact checkpoint SHA-256 is `f0b677d2cc028252002d15fac843ca7347b1b352eb2286a18bd202b724aa92b4`. The manuscript release manifest records hashes for the checkpoint, benchmark rows, training history, calibration report, empirical provenance, tables, figures, and PDF. Public-data source URLs and dataset-specific quality-control steps are stored in the empirical manifest.

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